EXATHON · TRACK 01 · MONEY, RISK & REGULATION

SparseMoney

An Intelligent Microsaving, Micro-Investment and Financial Reconciliation Platform

Every rupee tracked. Every mismatch explained.

TEAM - THE NERDS

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One platform. One transaction ID. One source of truth.

The Journey Begins with Everyday Transactions

A day of small spends (illustrative)

Food delivery	₹187
Shopping	₹243
Travel	₹10–₹100

Every saving, investment and report starts as a transaction record

What goes wrong without a system

Inconsistent saving

Saving depends on manual effort, so it is often skipped.

Unnoticed micro-spending

Small daily spends of ₹10–₹100 add up to large losses.

Investing feels out of reach

Seen as complex, risky and needing high capital.

Apps track, not guide

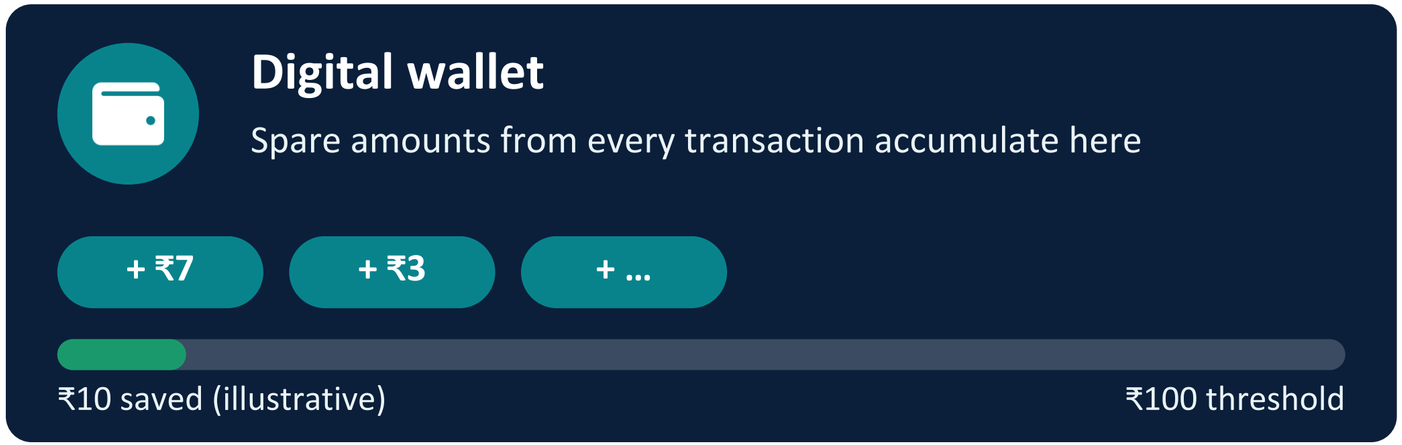
Expense apps show spending but do not change behaviour.

Why transaction-level visibility matters: if one record is wrong, everything built on it is wrong.

Microsaving: Small Transactions, Real Savings



₹250 – ₹243 = ₹7 saved automatically. Another example: ₹187 → ₹190 saves ₹3.



How it works

Automatic round-ups

Each purchase is rounded up and the spare amount is captured.

Smart wallet

Micro-savings are stored and tracked in one place.

No manual effort

Saving happens seamlessly, with no conscious decision.

Next: the wallet does not just hold money. At a set threshold it becomes an investment.

Micro-Investment: Every Small Amount Counts

Transaction
₹243 spent

→

Round-up
₹250, ₹7 spare

→

Wallet
Pool builds up

→

Threshold ₹100
Auto-invest fires

→

Invested
Order placed, logged

Where the pool goes

The rules engine picks the bucket, e.g. spare change above ₹25 goes all to gold.

- **Gold**
- **ETFs**
- **Index funds**
- **Debt funds**

Every order needs a record

- Backed by a matching wallet debit
- Failed or partial orders stay visible
- Each placement logged for audit

→ **Money now moves bank → wallet → investment.** Each hop must be recorded and verified.

Transaction Intelligence and Reconciliation

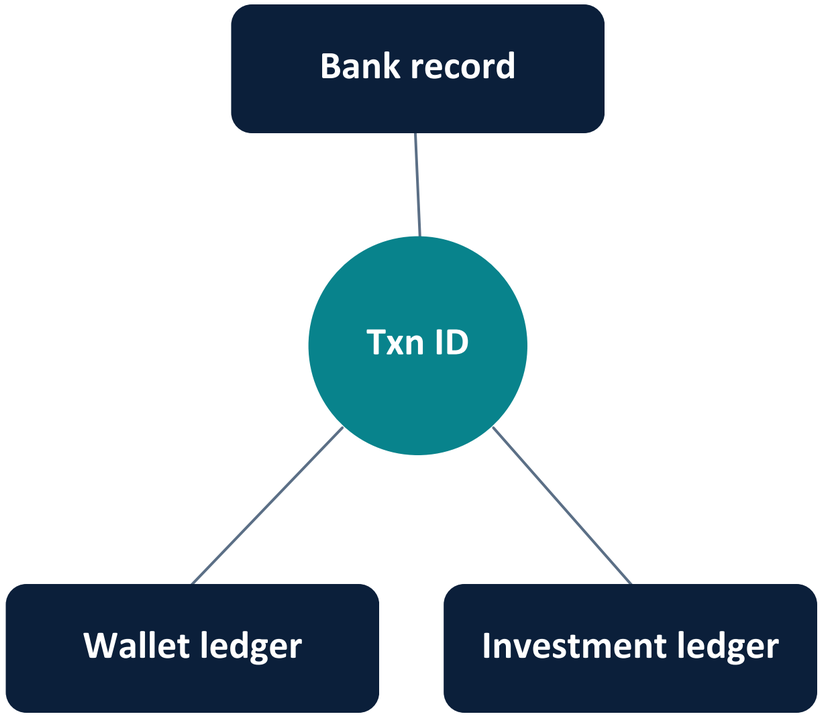
What is reconciliation?

Checking that the same transaction shows the same amount in every system that records it.

Txn	Engine / bank record	Wallet record	Result
Txn 1	₹3 spare	₹3 credited	✓ Matched
Txn 2	₹7 spare	— not found	Missing entry
Txn 3	Processed once	Processed twice	Duplicate
Txn 4	Expected amount	Different amount	Amount discrepancy

Illustrative records, matched on transaction ID

Three-way match



Why it matters: matching records keep savings and investment balances accurate.

Identifying Discrepancies and Financial Risks

Why records drift apart

- Delayed or duplicate bank webhooks
- Refunds and reversals after saving or investing
- Failed or partial investment orders
- Rounding and rule errors
- Retries that create duplicate entries

Transaction #TXN1042

HIGH RISK

Illustrative demo data, not actual financial records

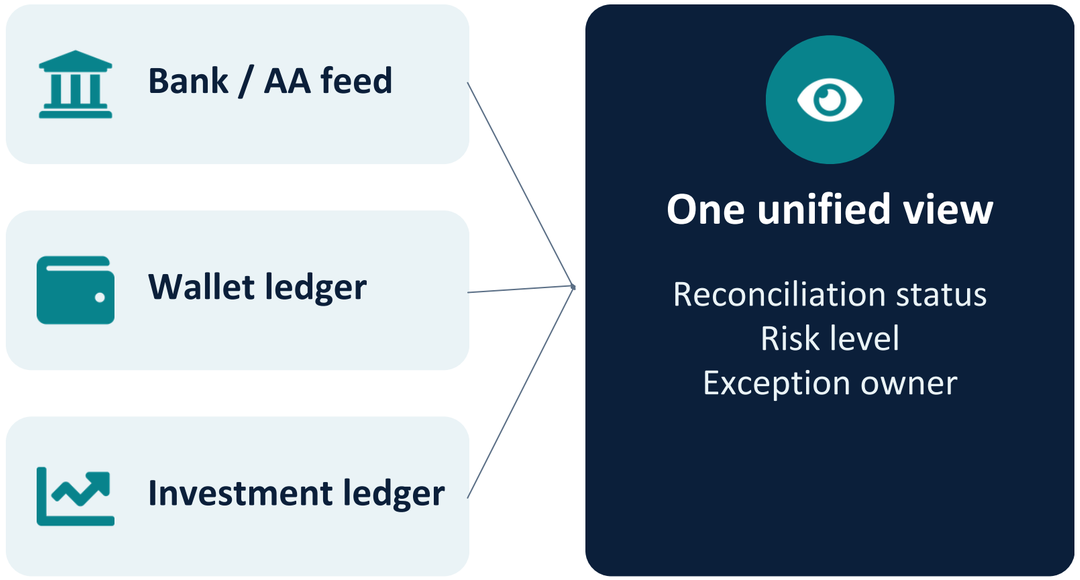
System	Recorded amount
Bank	₹243
Round-up engine	₹7
Wallet	**₹5**
Investment ledger	**₹0 (held)**

Rule R1 fired. Expected wallet credit ₹7, actual ₹5: a ₹2 mismatch.
Next step: hold the affected investment and send the transaction for investigation.

Left unresolved, mismatches mean wrong balances, wrong reports and hidden exposure.

Financial Monitoring and Risk Visibility

Systems that disagree, one view



Fragmented records blur decisions. One view shows exposure transaction by transaction.

Rule-based risk flags

Rule	What it detects	Default risk
R1	Amount mismatch	**Medium**
R2	Duplicate transaction ID	**High**
R3	Orphaned savings (reversal not mirrored)	**High**
R4	Unfunded investment order	**High**
R5	Unusual round-up pattern	**Low to Medium**
R6	Delayed match	**Medium**

R1 to R4 are in the prototype; R5 and R6 are designed for later.

Risk score = rule severity × amount × age of the exception. Each flag stores the rule, inputs and next action.

Transparent Decisions and Compliance

Every flag explains itself

RULE	R1: amount mismatch
NUMBERS	Expected ₹7, actual ₹5
NEXT STEP	Hold investment, investigate

The same pattern explains a held or declined action.

Immutable audit trail

Append-only, hash-chained events

Consent and data access

Consent and data-access records kept

Exact money

Integer paise; corrections never overwrite

Idempotency keys

A retried webhook cannot duplicate a round-up

IMPLEMENTED IN PROTOTYPE

- Rule flags R1 to R4 with explanations
- Held investments
- Exception queue and audit log

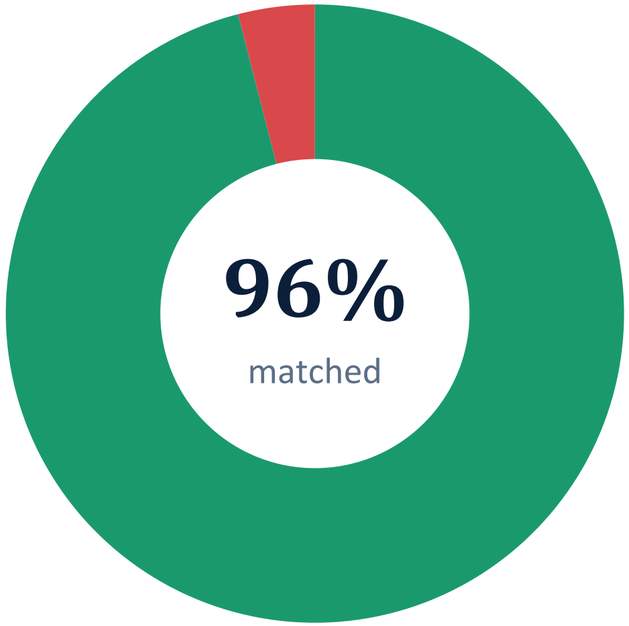
PROPOSED

- Plain-language customer notifications
- Rules R5 and R6

FUTURE OR OUT OF SCOPE

- LLM-phrased explanations
- Loan decisions (not covered)
- Certified compliance (needs regulated partners)

Automated Reconciliation and Reporting



96 matched, 4 unmatched. Illustrative demo data.

₹12,450

Reconciled amount

₹125

Discrepancy amount

Exception workflow

Pending → **Under Review** → **Resolved**

Every exception has an owner and resolution notes.

- Daily and monthly reconciliation reports
- Month-end close checklist
- Aged exception tracking

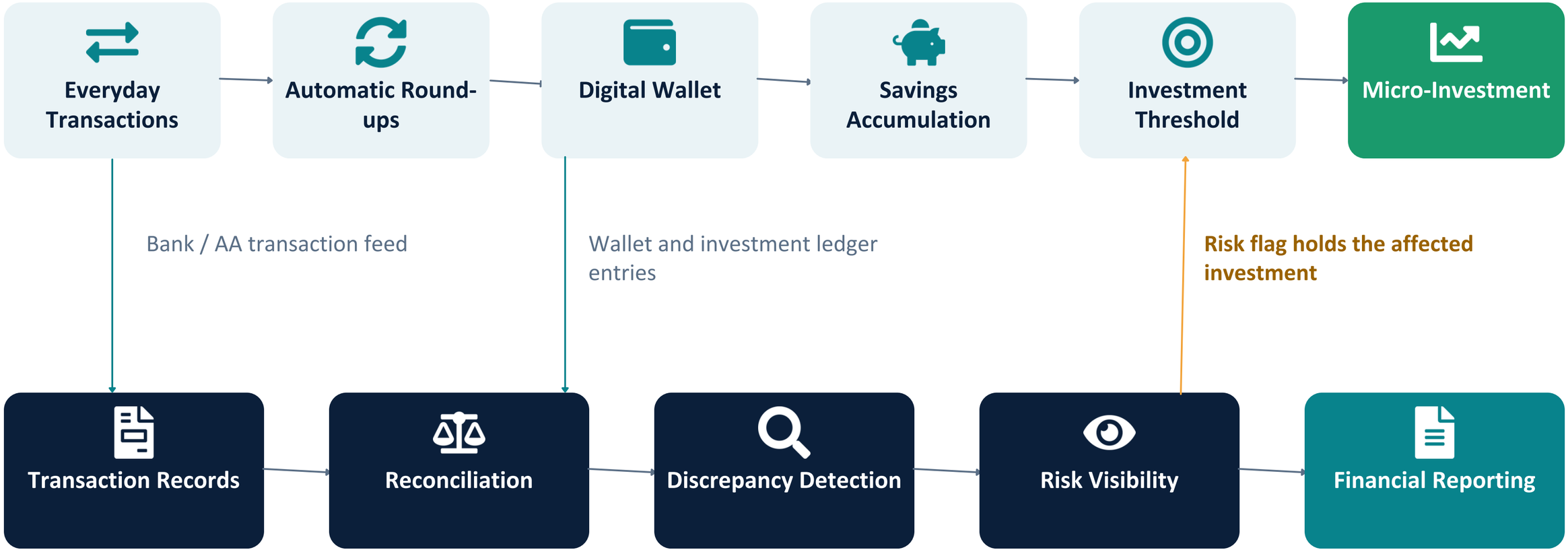
Metrics shown live

- **Match rate**
- **Time to detect a mismatch**
- **Time to resolve an exception**
- **Close time, before vs after**

Incremental runs: only new or changed records are rechecked each cycle, so work does not pile up.

SparseMoney: One Integrated Workflow

SAVE AND INVEST



One transaction ID links both flows, so every rupee can be traced, matched and explained.

Technology, Innovation and Track Alignment

Architecture and stack

Experience	React + TypeScript, Tailwind CSS, Recharts, Framer Motion. Customer and Operations dashboards
Services	Node.js + Express with idempotent ingestion. Round-up engine, reconciliation worker, rule engine
Data	Supabase (Postgres): append-only ledgers, exceptions, hash-chained audit events, amounts in paise
Simulation	Mock bank and Account Aggregator feed with fault injection. Simulated investment orders

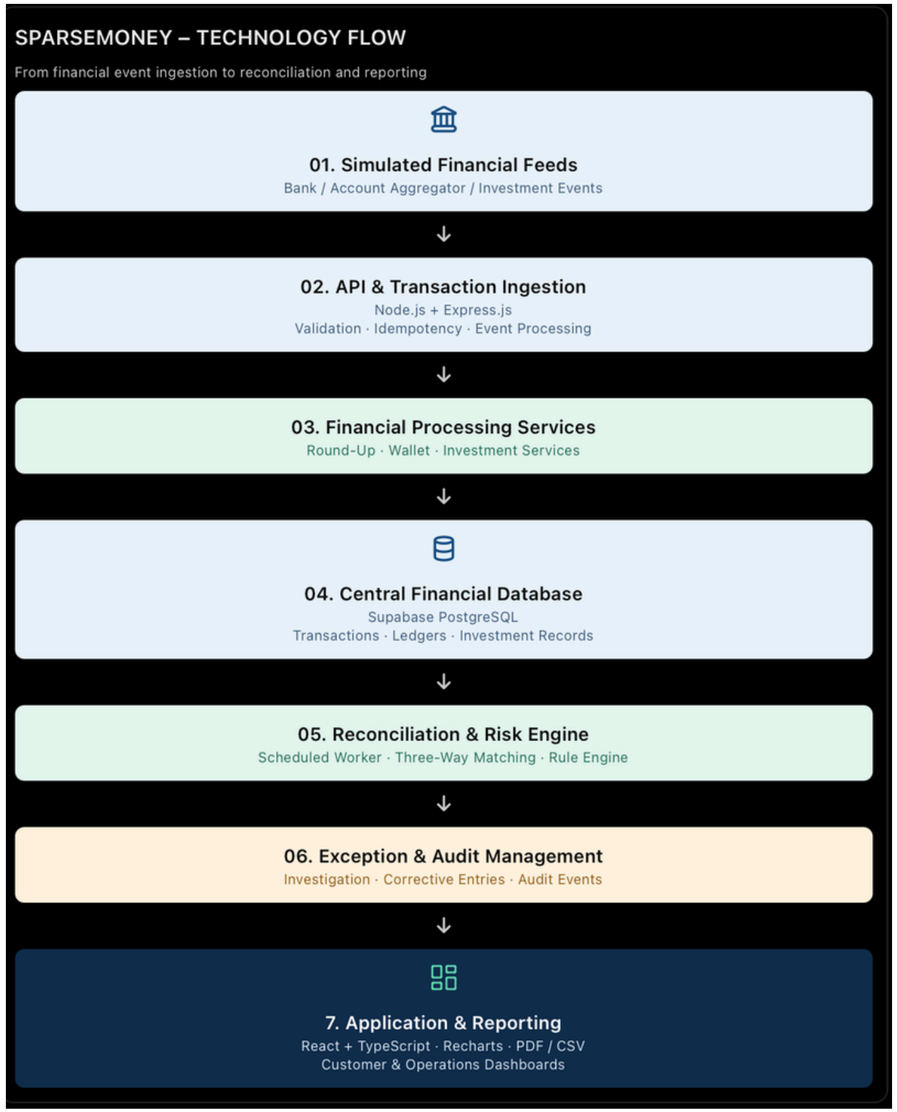
INNOVATION

Three-way matching on one transaction ID

Every flag names rule, numbers, next step

Idempotency keys stop duplicate round-ups

Integer paise and append-only entries



Impact, Future Scope and Conclusion

Accessible savings

Micro-investment

Financial accuracy

Reconciliation

Financial intelligence

Benefits for users

- Save and invest without extra effort
- Investing made simple for beginners
- Clear status and plain-language issue alerts

Benefits for institutions

- One view across bank, wallet and investment records
- Mismatches caught early and traceable
- Less manual reconciliation, simpler close

Future scope

- Group saving and Horizon Predictor
- Spending insights engine
- Real AA and broker integrations via regulated partners
- LLM-phrased customer explanations

Honest limits: simulated bank, AA and broker feeds · rule-based detection, not ML · not a certified compliance product

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Thank you Team The Nerds: Lakshmi M | Litheswar M | Madhan V M